

# RAYAT BAHRA PROFESSIONAL UNIVERSITY



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NAME OF SUBJECT- WEB DESIGNING

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SEMESTER-I

## **Unit B -Introduction to HTML and its Elements**

**Topic II-** Introduction to elements of HTML;  
Working with Text; Working with Lists, Tables and  
Frames; Working with Hyperlinks, Images and  
Multimedia; Working with Forms and controls.

# Introduction to elements of HTML



- **HTML elements are building block** of a web page.
- It tells the browser **how to display content**.

Structure of an HTML Element:

`<opening-tag> Content </closing-tag>`

**Opening tag** → starts the element (e.g., `<p>`)

**Content** → the actual text or data inside (e.g., Hello)

**Closing tag** → ends the element (e.g., `</p>`)

`<p>Hello</p>`

This is a **paragraph element**.

# Types of HTML Elements



## 1. Text Elements

- `<h1>` to `<h6>` → Headings
- `<p>` → Paragraph
- `<b>`, `<i>`, `<u>` → Bold, Italic, Underline
- `<br>` → Line Break (empty element, no closing tag)

## 2. Image & Multimedia Elements

- `<img>` → Image
- `<audio>` → Audio player
- `<video>` → Video player



### **3.List Elements**

- `<ul>` → Unordered list
- `<ol>` → Ordered list
- `<li>` → List item

### **4.Table Elements**

- `<table>` → Table
- `<tr>` → Table row
- `<td>` → Table cell
- `<th>` → Table header



## 5. Form Elements

- `<form>` → Form container
- `<input>` → Input field
- `<textarea>` → Multi-line text box
- `<select>` and `<option>` → Dropdown menu
- `<button>` → Button

## 6. Structural Elements

- `<div>` → Division / section
- `<span>` → Inline container
- `<header>`, `<footer>`, `<nav>`, `<section>`, `<article>` → Semantic structure

# EXAMPLE



```
<h1>This is a Heading</h1>
```

```
<p>This is a paragraph with <b>bold text</b>.</p>
```

```

```

```
<ul>
```

```
<li>HTML</li>
```

```
<li>CSS</li>
```

```
<li>JavaScript</li>
```

```
</ul>
```

This uses multiple **HTML elements**: heading, paragraph, bold text, image, and list.

# Frames



- HTML frames are used to divide your browser window into multiple sections where each section can load a separate HTML document independently.
- A collection of frames in the browser window is known as a frameset.
- The window is divided into frames in a similar way the tables are organized: into rows and columns.

# Creating Frames in HTML



- To make frames on a page we use **<frameset>** tag instead of **<body>** tag.
- The **<frameset>** tag defines how to divide the window into frames.
- The **rows** attribute of **<frameset>** tag defines horizontal frames and **cols** attribute defines vertical frames.
- Each frame is indicated by **<frame>** tag and it defines which HTML document shall open into the frame.

# Attributes of frameset Tag

| cols         | <p>Specifies how many columns are contained in the frameset and the size of each column. For example to create three vertical frames, use <i>cols="10%, 80%,10%"</i>.</p> <p>Using a wildcard symbol. For example to create three vertical frames, use <i>cols="10%, 10%,*"</i>. In this case wildcard takes remainder of the window.</p> |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rows         | <p>It is used to specify the rows in the frameset.</p> <p>For example to create two horizontal frames, use <i>rows="10%, 90%"</i>.</p>                                                                                                                                                                                                    |
| border       | <p>This attribute specifies the width of the border of each frame in pixels. For example <i>border="5"</i>. A value of zero means no border.</p>                                                                                                                                                                                          |
| frameborder  | <p>This attribute specifies whether a three-dimensional border should be displayed between frames. This attribute takes value either 1 (yes) or 0 (no). For example <i>frameborder="0"</i> specifies no border.</p>                                                                                                                       |
| framespacing | <p>This attribute specifies the amount of space between frames in a frameset. This can take any integer value. For example <i>framespacing="10"</i> means there should be 10 pixels spacing between each frames.</p>                                                                                                                      |

# HTML <frame> Tag Attributes

| src          | This attribute is used to give the file name that should be loaded in the frame. Its value can be any URL. For example, src="abc.html"                                                               |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| name         | This attribute allows you to give a name to a frame. It is used to indicate which frame a document should be loaded into.                                                                            |
| marginwidth  | This attribute allows you to specify the width of the space between the left and right of the frame's borders and the frame's content. The value is given in pixels. For example marginwidth="10".   |
| marginheight | This attribute allows you to specify the height of the space between the top and bottom of the frame's borders and its contents. The value is given in pixels. For example marginheight="10".        |
| noresize     | By default you can resize any frame by clicking and dragging on the borders of a frame. The noresize attribute prevents a user from being able to resize the frame. For example noresize="noresize". |
| scrolling    | This attribute controls the appearance of the scrollbars that appear on the frame. This takes values either "yes", "no" or "auto". For example scrolling="no" means it should not have scroll bars.  |

# Disadvantages of Frames

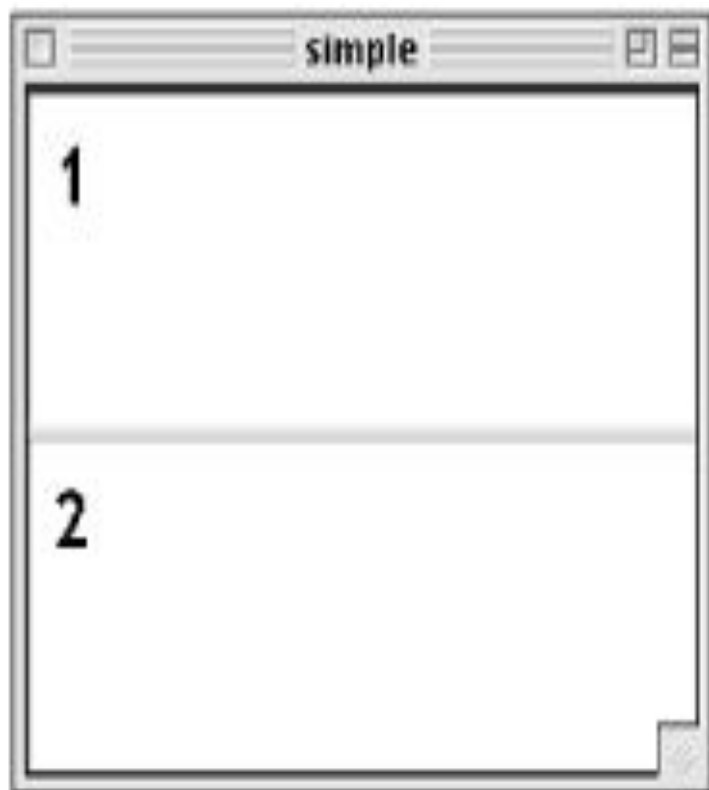


- Some smaller devices cannot cope with frames often because their screen is not big enough to be divided up.
- Sometimes your page will be displayed differently on different computers due to different screen resolution.
- The browser's *back button* might not work as the user hopes.
- There are still few browsers that do not support frame technology.

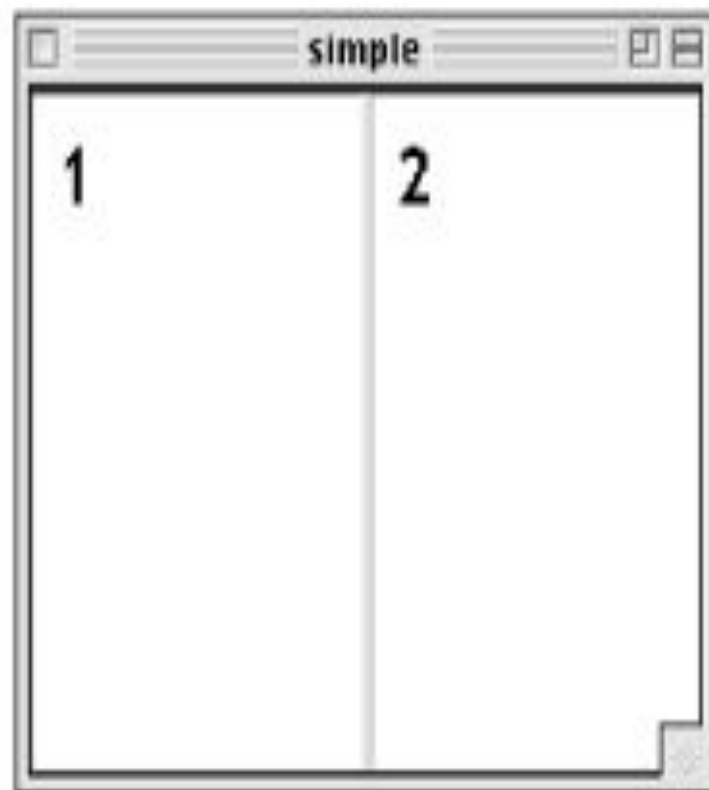
## Note



- The <frame> tag is no longer recommended as it is not supported by HTML5. Instead of using this tag, we can use the <iframe> or <div> with CSS to achieve the similar effects.



```
<FRAMESET ROWS="*,*">  
  <FRAME SRC="1.html">  
  <FRAME SRC="2.html">  
</FRAMESET>
```



```
<FRAMESET COLS="*,*">  
  <FRAME SRC="1.html">  
  <FRAME SRC="2.html">  
</FRAMESET>
```

# SIMPLE CODE



```
<html>
<head><title>frames</title></head>
<frameset rows="50,50">
  <frameset cols="50%,50%">
    <frame src="audio.html" >
      <frame src=video.html >
  </frameset>
  <frameset cols="50%,50%">
    <frame src="table.html">
    <frame src="map.html">
  </frameset>
</frameset>
</html>
```

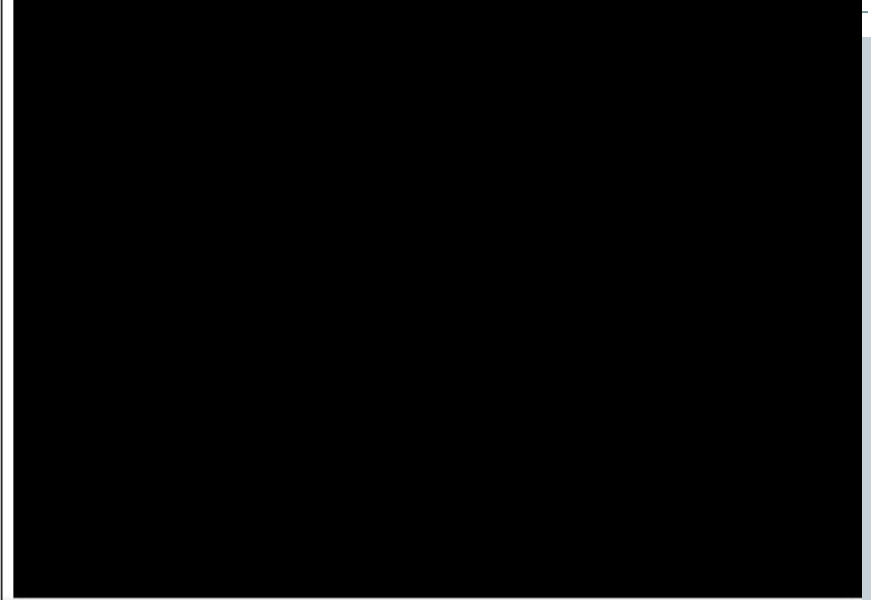
# CODE



```
<html>
<head><title>frames</title></head>
<frameset rows="50,50">
  <frameset cols="50%,50%">
    <frame src="audio.html" scrolling=yes marginheight="10"
marginwidth="50">
    <frame src=video.html noresize>
  </frameset>
  <frameset cols="50%,50%">
    <frame src="table.html" noresize>
    <frame src="map.html" scrolling=auto noresize>
  </frameset>
</frameset>
</html>
```

# OUTPUT

▶ 0:00 / 0:24 — 🔊 ⋮



| LIST | MARKS |       |
|------|-------|-------|
|      | NAME  | AGE   |
|      | shivi | shiv  |
|      | neha  | mahak |

Personal Information



# For Practice

|            |            |            |
|------------|------------|------------|
| Frame<br>1 | Frame<br>2 | Frame<br>3 |
| Frame<br>4 | Frame<br>5 | Frame<br>6 |

## Great Recipes

### CONTENTS

[Breakfast  
Tacos](#)

[Greek Salad](#)

[Mexican Beans  
and Rice](#)

[Refried Beans](#)

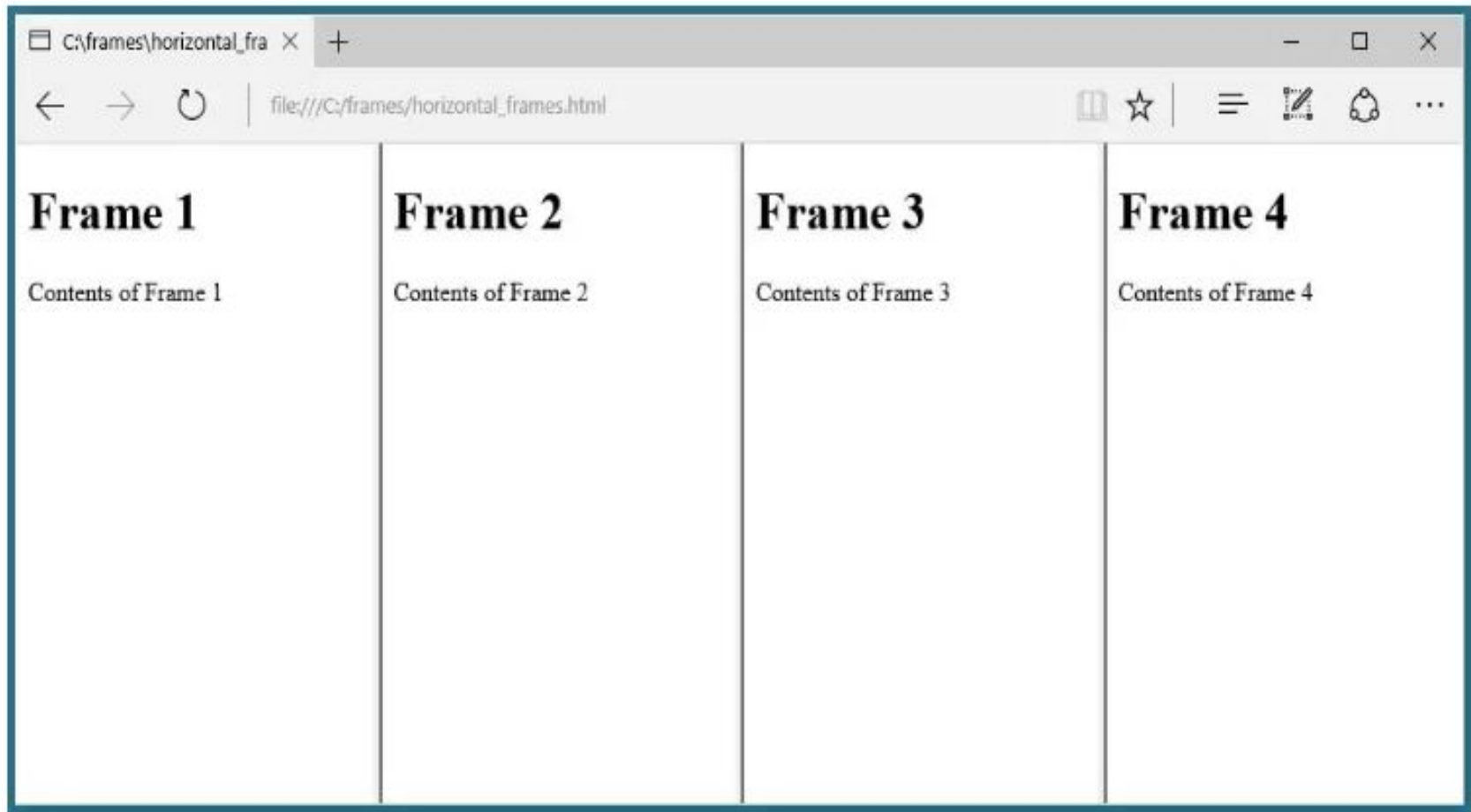
[Spanish Rice](#)

[Sweet & Sour  
Bean Dish](#)

### Breakfast Tacos

- two good sized red potatoes
- small onion
- bell pepper
- green onions
- dash of: cumin, chili powder, garlic, oregano, basil, pepper
- pinch of salt
- fresh cilantro
- bit of oil
- tomatoes
- whole wheat tortillas
- salsa (optional)
- lime (optional)

# For Practice



# Multimedia



Multimedia comes in many different formats. It can be almost anything you can hear or see, like images, music, sound, videos, records, films, animations, and more.

## Types of Multimedia Content for HTML

1. Images:
2. Audio
3. Video
4. Animations:

# Common Video Formats



|             |                       |                                                                                                                                                                 |
|-------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MPEG</b> | <b>.mpg<br/>.mpeg</b> | <b>MPEG. Developed by the Moving Pictures Expert Group. The first popular video format on the web. Not supported anymore in HTML.</b>                           |
| AVI         | .avi                  | AVI (Audio Video Interleave). Developed by Microsoft. Commonly used in video cameras and TV hardware. Plays well on Windows computers, but not in web browsers. |
| WMV         | .wmv                  | WMV (Windows Media Video). Developed by Microsoft. Commonly used in video cameras and TV hardware. Plays well on Windows computers, but not in web browsers.    |
| QuickTime   | .mov                  | QuickTime. Developed by Apple. Commonly used in video cameras and TV hardware. Plays well on Apple computers, but not in web browsers.                          |
| RealVideo   | .rm<br>.ram           | RealVideo. Developed by Real Media to allow video streaming with low bandwidths. Does not play in web browsers.                                                 |



| Format           | File         | Description                                                                                                                                               |
|------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Flash            | .swf<br>.flv | Flash. Developed by Macromedia. Often requires an extra component (plug-in) to play in web browsers.                                                      |
| Ogg              | .ogg         | Theora Ogg. Developed by the Xiph.Org Foundation. Supported by HTML.                                                                                      |
| WebM             | .webm        | WebM. Developed by Mozilla, Opera, Adobe, and Google. Supported by HTML.                                                                                  |
| MPEG-4<br>or MP4 | .mp4         | MP4. Developed by the Moving Pictures Expert Group. Commonly used in video cameras and TV hardware. Supported by all browsers and recommended by YouTube. |

# Common Audio Formats



| Format    | File          | Description                                                                                                                                                                                                                                                                                    |
|-----------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MIDI      | .mid<br>.midi | MIDI (Musical Instrument Digital Interface). Main format for all electronic music devices like synthesizers and PC sound cards. MIDI files do not contain sound, but digital notes that can be played by electronics. Plays well on all computers and music hardware, but not in web browsers. |
| RealAudio | .rm<br>.ram   | RealAudio. Developed by Real Media to allow streaming of audio with low bandwidths. Does not play in web browsers.                                                                                                                                                                             |
| WMA       | .wma          | WMA (Windows Media Audio). Developed by Microsoft. Plays well on Windows computers, but not in web browsers.                                                                                                                                                                                   |
| AAC       | .aac          | AAC (Advanced Audio Coding). Developed by Apple as the default format for iTunes. Plays well on Apple computers, but not in web browsers.                                                                                                                                                      |



| Form<br>at | File | Description                                                                                                                                                                                  |
|------------|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ogg        | .ogg | Ogg. Developed by the Xiph.Org Foundation. Supported by HTML.                                                                                                                                |
| MP3        | .mp3 | MP3 files are actually the sound part of MPEG files. MP3 is the most popular format for music players. Combines good compression (small files) with high quality. Supported by all browsers. |
| MP4        | .mp4 | MP4 is a video format, but can also be used for audio. Supported by all browsers.                                                                                                            |
| WAV        | .wav | WAV. Developed by IBM and Microsoft. Plays well on Windows, Macintosh, and Linux operating systems. Supported by HTML.                                                                       |

# Audio Attribute Specification:

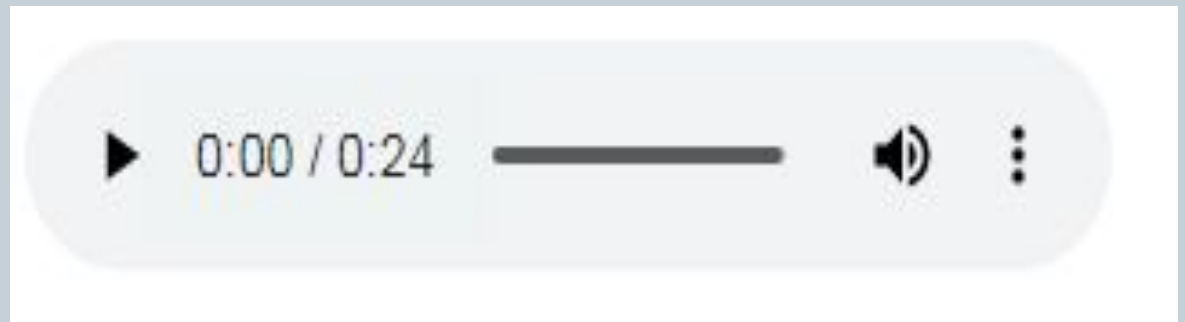
- The HTML5 audio tag can have a number of attributes to control the look and feel and various functionalities of the control:

| Attribute | Description                                                                                                                                               |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| autoplay  | This boolean attribute if specified, the audio will automatically begin to play back as soon as it can do so without stopping to finish loading the data. |
| controls  | If this attribute is present, it will allow the user to control audio playback, including volume, seeking, and pause/resume playback.                     |
| loop      | This boolean attribute if specified, will allow audio automatically seek back to the start after reaching at the end.                                     |
| preload   | This attribute specifies that the audio will be loaded at page load, and ready to run. Ignored if autoplay is present.                                    |
| src       | The URL of the audio to embed. This is optional; you may instead use the <source> element within the video block to specify the video to embed            |

# CODE



```
<html>  
<head><title>audio</title></head>  
<body>  
<audio src="S b k neika.mp3" controls loop autoplay  
></audio>  
</body>  
</html>
```



# Video Attribute Specification:



- The HTML5 video tag can have a number of attributes to control the look and feel and various functionalities of the control:

| Attribute | Description                                                                                                                                               |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| autoplay  | This boolean attribute if specified, the video will automatically begin to play back as soon as it can do so without stopping to finish loading the data. |
| controls  | If this attribute is present, it will allow the user to control video playback, including volume, seeking, and pause/resume playback.                     |
| height    | This attribut specifies the height of the video's display area, in CSS pixels.                                                                            |
| loop      | This boolean attribute if specified, will allow video automatically seek back to the start after reaching at the end.                                     |
| src       | The URL of the video to embed. This is optional; you may instead use the <source> element within the video block to specify the video to embed            |
| width     | This attribut specifies the width of the video's display area, in CSS pixels.                                                                             |

# CODE



```
<html>  
<body><video width="640" height="360"  
  src="VID-20141203-WA0006.mp4" autoplay  
  controls loop ></video>  
</body>  
</html>
```



# Playing a YouTube Video in HTML



- To play your video on a web page, do the following:
- Upload the video to YouTube
- Take a note of the video id
- Define an `<iframe>` element in your web page
- Let the `src` attribute point to the video URL
- Use the `width` and `height` attributes to specify the dimension of the player



```
<html>
<body>
<iframe width="420" height="345"
  src="https://www.youtube.com/embed/tgbNymZ7vq
  Y">
</iframe>
</body>
</html>
```

